# Machine Learning — Homework 1

Representation, Geometry, and Completion

*Chapters 1–4*

Choose a numerical dataset and turn it into a *completion problem.*

Some information about each object must be regarded as observed and some as hidden. Your goal is to reconstruct the hidden information from what remains visible.

The problem is deliberately underspecified. Choosing a sensible representation, observation rule, geometry, method, and validation procedure is part of the homework.

## Task

Produce an executable notebook in which you:

1. define the objects and their matrix representation;
2. define what is observed and what is hidden;
3. construct a simple baseline completion rule;
4. construct a linear completion method using appropriate numerical or machine-learning libraries;
5. investigate a low-dimensional representation of the data using PCA, SVD, or an equivalent method;
6. use the low-dimensional structure to attempt completion of the hidden information;
7. change the observation pattern in a meaningful way and determine what changes in the result.

Your experiments should contain enough numerical or graphical evidence to support the conclusions you draw.

## One required check

For a centered data matrix $X_c$, choose a rank $r$ and construct the corresponding truncated SVD

$$X_r = U_r \Sigma_r V_r^\top .$$

Form the projectors

$$P_r^{(s)} = U_r U_r^\top , \qquad P_r^{(f)} = V_r V_r^\top ,$$

and verify numerically that

$$P_r^{(s)} X_c \approx X_c P_r^{(f)} \approx P_r^{(s)} X_c P_r^{(f)} \approx X_r .$$

Briefly explain why this does not imply that either projector is the identity.

## Use of AI

You may use an AI system freely for coding, debugging, mathematical clarification, or experimental design.

Include, as an appendix to the final report, three short examples of interactions with the AI that influenced your solution. For each one, state:

- what you asked;
- what choice or assumption was involved;
- whether you accepted or modified the answer.

Do not submit a complete chat transcript.

## Submission

Submit:

- one notebook that runs from beginning to end, commented when necessary.
- a short but effective report, that summarizes crucial steps and results in the notebook. The conclusion should explicitly address the question:

  *What information or structure made the hidden values predictable?*

There is no unique correct pipeline. The quality of the work will be judged from the coherence of the choices (from the very first one), the correctness of the implementation, and the ability to interpret what the methods are doing.

## Grading

Grading is arbitrary, mainly based on the immediate superficial impression from reading the report. Your instructor likes clarity in problems and solutions presentation, conciseness, to-the-point observations, and doesn't like reading code. The grade [1..100] points will be inverse proportional to the amount of work needed to appreciate your work.